Department of Computer Science

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Digital Forensics and Investigations

COS783

Practical 1

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Introduction

This investigation examined a forensic image from a university-issued Windows 10 device suspected of policy violations. The objective was to identify traces of unauthorized file sharing, malware execution, and anonymized browsing.

Scope

The process involved verifying the image's integrity, extracting timeline and browser artifacts, identifying suspicious software, and analyzing persistence mechanisms and deleted data. All procedures were conducted using approved forensic tools within a secure, controlled environment.

Methodology

The forensic process began with validating the integrity of the acquired image using SHA-256 hashing. The image was then mounted in a controlled analysis environment configured with Autopsy and PowerShell. Timeline analysis was conducted to reconstruct user activity, focusing on browser history, file access patterns, and system events. Artifacts related to anonymized browsing, torrent usage, and executable files were extracted and examined. Registry entries, startup configurations, and deleted file traces were analyzed to identify persistence mechanisms and concealment strategies. All findings were documented with timestamps and verified against system logs to ensure accuracy and reproducibility.

Task 1: Preparation and Verification of Image Integrity

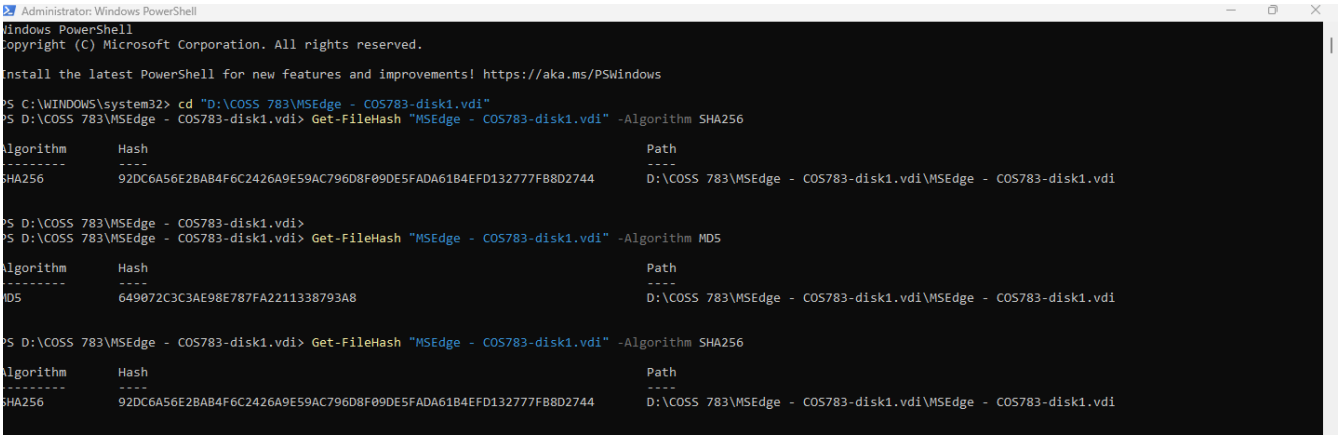
1.1. Hashes of the forensic image provided using both md5sum and sha256sum

md5sum command

```
Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm MD5
```

sha256 command

```
Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd "D:\COSS 783\MSEdge - COS783-disk1.vdi"
PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          92DC6A56E2BAB4F6C2426A9E59AC796D8F09DE5FADA61B4EFD132777FB8D2744       D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi


PS D:\COSS 783\MSEdge - COS783-disk1.vdi>
PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm MD5

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             649072C3C3AE98E787FA2211338793A8                                       D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi


PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          92DC6A56E2BAB4F6C2426A9E59AC796D8F09DE5FADA61B4EFD132777FB8D2744       D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi
```

Figure 1: md5sum and sha256sum output on Powershell

1.2. Importance of hashing in digital forensics

1. **Evidence Integrity**
   Hashing creates a unique digital signature to confirm that forensic data remains unchanged during acquisition, analysis, or transfer.

2. **Chain of Custody**
   Tracking hash values at each stage ensures a verifiable custody trail, proving the evidence is identical to its original form for legal use.

1.3. Image Integrity

The forensic image was verified using the `sha256sum` utility in PowerShell. The resulting hash value was:

`92dc6a56e2bab4f6c2426a9e59ac796d8f09de5fada61b4efd132777fb8d2744`

Hashing is essential in digital forensics as it ensures the integrity of evidence by producing a unique digital fingerprint. Any alteration to the file would result in a different hash, thereby alerting investigators to potential tampering. The image's integrity was confirmed, and no discrepancies were found.

Task 2: Forensic Acquisition

2.1. Loading the image on Autopsy(the chosen tool)

The **MSEdge - COS783-disk1.vdi** was converted to the raw format, and then the raw image was ingested instead. Autopsy handles raw images much more

consistently.

Conversion Command

```
VBoxManage clonehd "D:\COSS 783\MSEdge -
COS783-disk1.vdi\MSEdge - COS783-disk1.vdi" "D:\COSS
783\MSEdge.raw" --format RAW
```

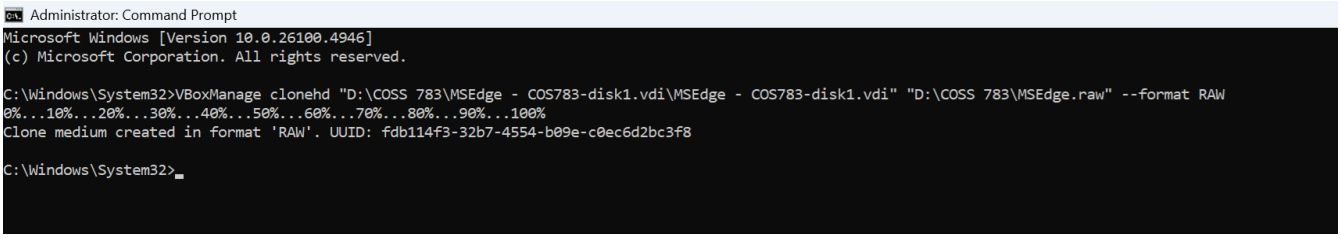


Figure 2: Successful .raw file saved to COSS 783 folder

SHA256 hashes were computed for both the .raw and .vdi disk images. The differing hash values on the following console window confirm that the files are structurally distinct, validating the integrity of each format for its respective forensic purpose.

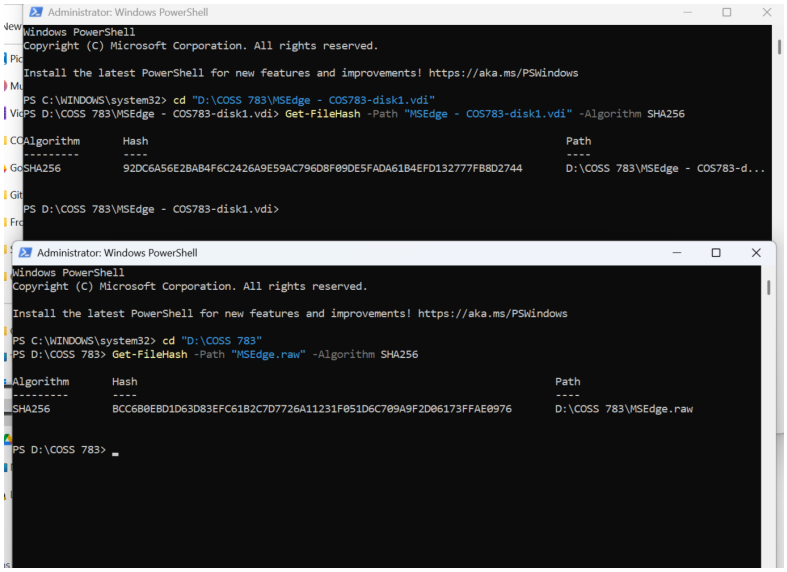


Figure 3: .raw and .vdi disk image hashes

Data source

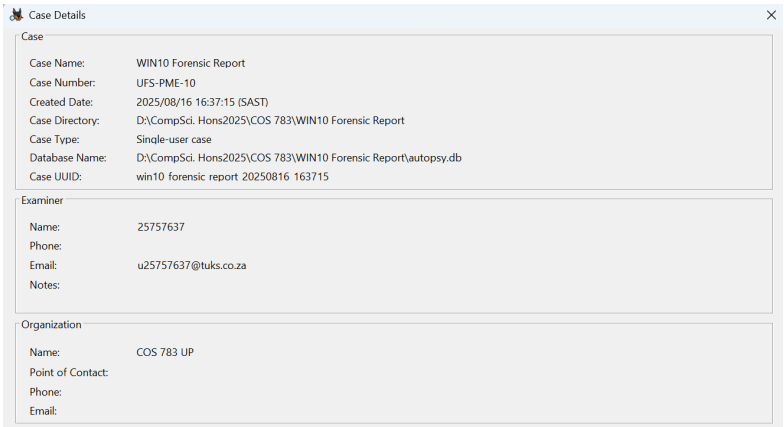


Figure 4: Digital Forensics Case Details

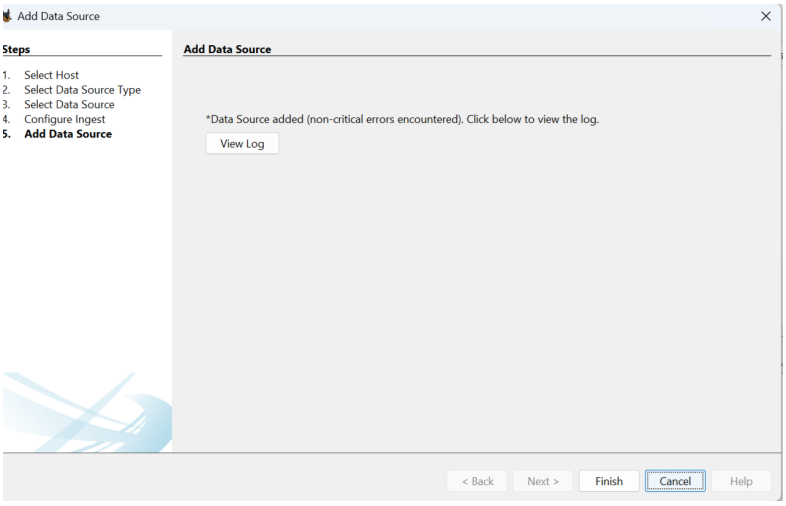


Figure 5: Successful data source loading

After creating a new case on Autopsy and adding the data source, MSEdge.raw, the following ingest modules were enabled:

- Recent Activity
- Operating System Information
- File Type Identification

- Extension Mismatch Detector
- Embedded File Extractor
- Keyword Search
- Email Parser
- Encryption Detection
- Interesting Files Identifier
- Virtual Machine Extractor
- Data Source Integrity

2.2. System Metadata

a) Hostname: `MSEDGEWIN10`

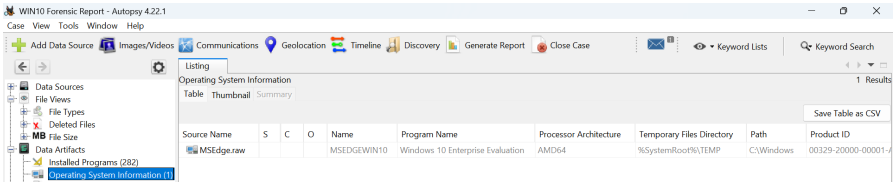


Figure 6: Data Artifact = OS Information

b) User Accounts

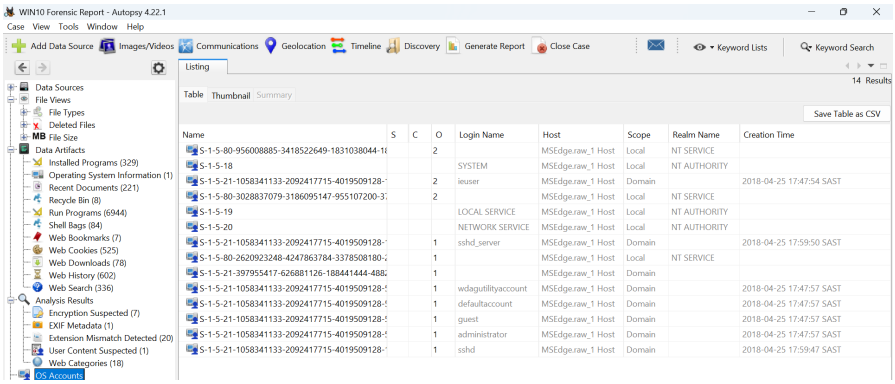


Figure 7: OS Accounts

c) Last Shutdown Time

2.3 File system and Partition structure

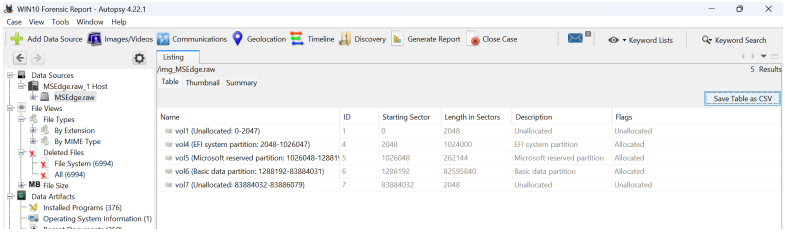


Figure 8: Output obtained from the disk image

Partitioning Scheme: GPT (based on presence of EFI and Microsoft reserved partitions)
Total Partitions: 5

- vol1: Unallocated (sectors 0–2047)
- vol4: EFI System Partition (sectors 2048–1026047)
- vol5: Microsoft Reserved Partition (sectors 1026048–1288191)
- vol6: Basic Data Partition (sectors 1288192–83884031)
- vol7: Unallocated (sectors 83884032–83886079)

Primary File System: NTFS (used in vol6, the main Windows partition)

Interpretation: The disk image uses a GPT layout with standard Windows partitions. The NTFS file system is used for the main data volume, typical of Windows 10 installations.

Task 3: Timeline Analysis and User Activity

3.1. When the suspicious activity started

The following are the filters used on the timeline to search for suspicious activity:

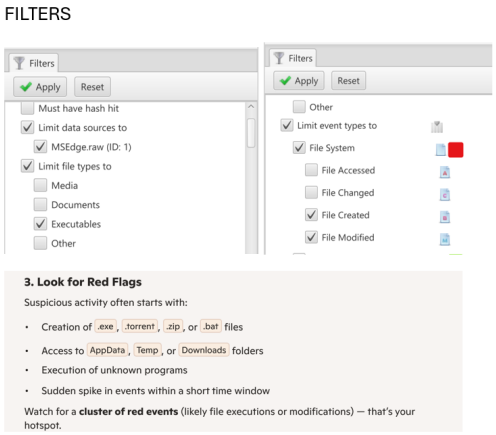


Figure 9: Filters

Suspicious Activity Timeline

Date	Event	Evidence	Interpretation
2018-01-25	Execution of `sdelete.exe`	Found in `2018 Jan.csv`	First sign of anti-forensics; secure deletion tool often used to erase traces
2018-02-13	VirtualBox Guest Additions installed	Found in `2018 Feb.csv`	Suggests VM setup — could be attacker sandbox or analyst environment
2018-08-30	Malware indicators in DLLs and cache files	`mpengine.dll`, `mpasdesc.dll.mui`, `f0dbba63ec90be3`	TrojanDownloader, Worm, Backdoor behavior; possible data staging or exfiltration
2018 (undated)	Legacy registry artifact	`ntuser.dat.LOG1`	May reveal historical user behavior or software installs

2025-07-30 11:08:49	Chrome extension with “eraser” keyword	Located in `component_crx_cache`	Suggests privacy-focused or wiping extension; identical timestamps — possible anti-forensic intent
2025-07-30 12:16:40	Deletion of `utorrent_-installer.exe`	Recycle Bin artifact `$R00KXBM.exe`	Manual deletion linked to torrent activity; user attempted concealment without secure wipe
2025-07-30	CCleaner keyword in Avast directory	Timestamps zeroed	Indicates prior use or detection of anti-forensic tool; metadata wiping suspected
2025-07-30	Access to Chrome Secure Preferences & Local State	Browser config files	Confirms browser usage; may contain encrypted traces of extensions or privacy tools
2025-07-30	Coordinated DLL and cache file modifications	Found in `2025 July18.csv` & `July18b.csv`	Malware reactivation or persistence; mirrors 2018 indicators

”worm” key search

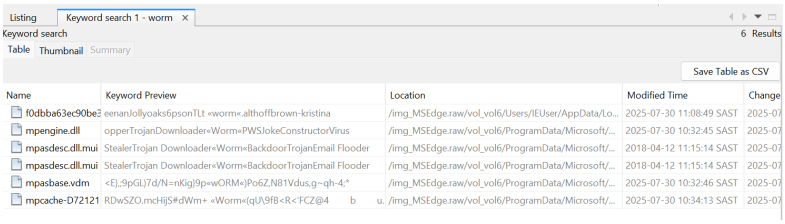


Figure 10: ”worm” search output

3.2. File downloads (e.g., .torrent or suspicious executables)

- **Torrent Activity:** Multiple `.torrent` files from YTS.AG and YTS.MX suggest active media downloads.
- **Software Acquisition:** Transmission installer (`.msi`) and Ubuntu ISOs indicate software-related downloads, possibly for dual-booting or forensic evasion.

- **Zone.Identifier Tags:** All downloads tagged with Zone ID 3, confirming internet origin.
- **Suspicious File:** Obfuscated filename in `/Videos` may represent encrypted payload or container.

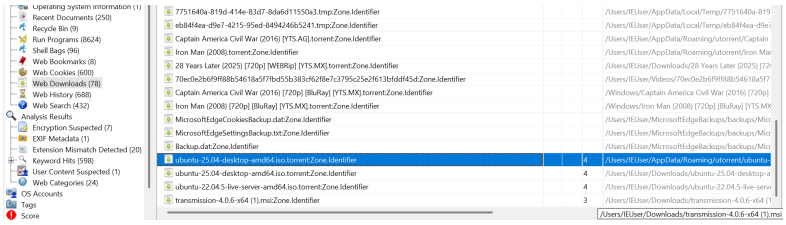


Figure 11: Web Downloads Output

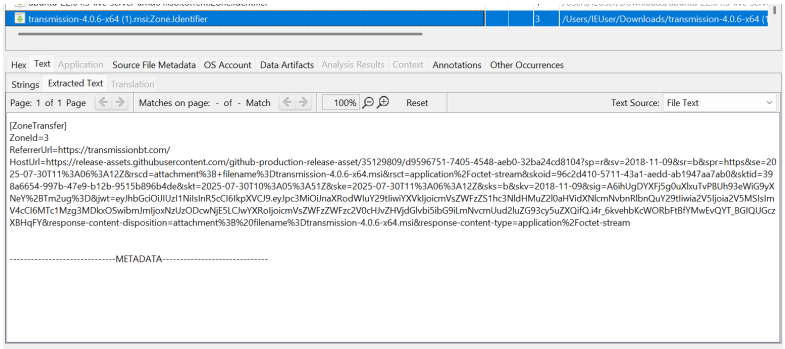


Figure 12: Analysis of web downloads zoneID

3.3. Any evidence of file deletion or anti-forensic behaviour

- **Recycle Bin Artifact:** `utorrent_installer.exe` manually deleted on 2025-07-30.
- **CCleaner Keyword:** Found in `/ProgramData/Avast Software/` with zeroed timestamps — possible metadata wiping.
- **Eraser Keyword:** Located in Chrome extension cache; identical timestamps suggest single-use install or timestamp manipulation.
- **Browser Artifacts:** Secure Preferences and Local State files accessed on 2025-07-30 — may contain encrypted traces of user activity.

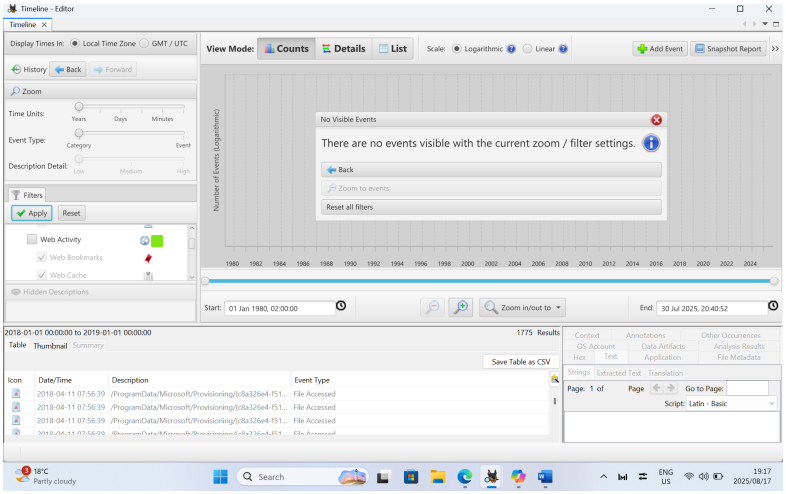


Figure 13: Output of deleted files using filters on the timeline

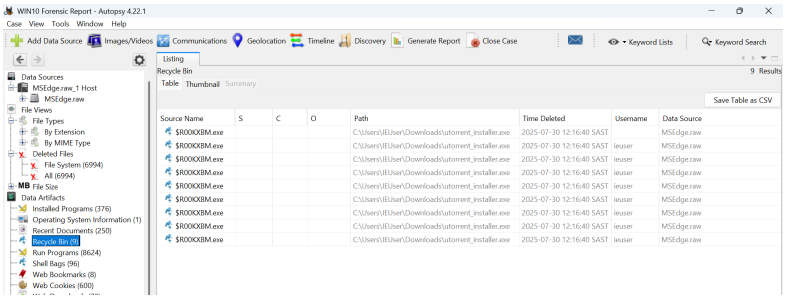


Figure 14: Recycle Bin Output

”CCleaner” Key Search

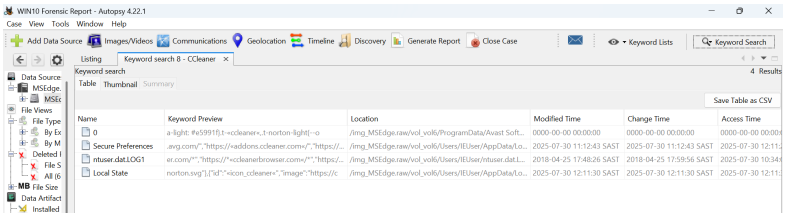


Figure 15: ”CCleaner” Key Search Output1

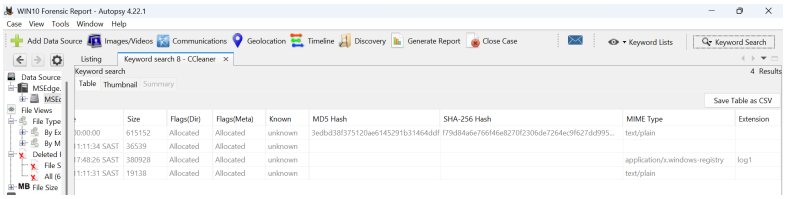


Figure 16: ”CCleaner” Key Search Output2

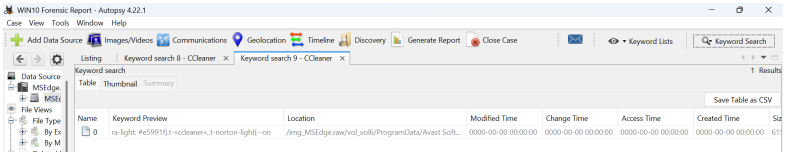


Figure 17: ”CCleaner” Key Search Output3

”secure delete” Key Search

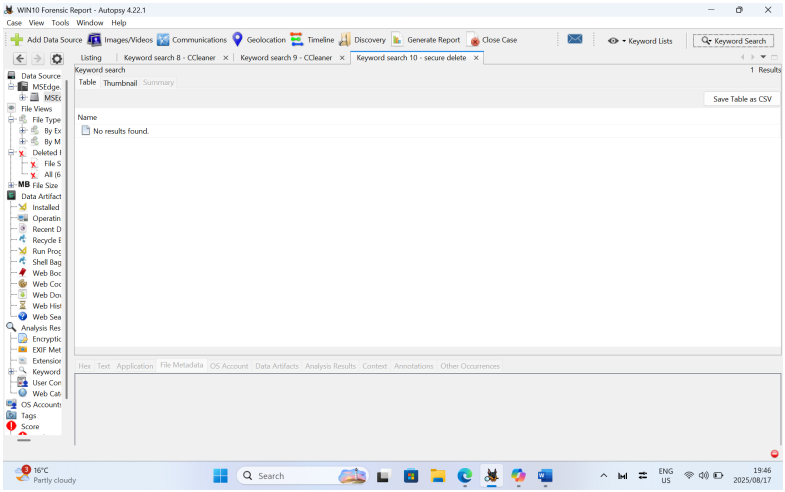


Figure 18: ”secure delete” Key Search Output

”eraser” key search

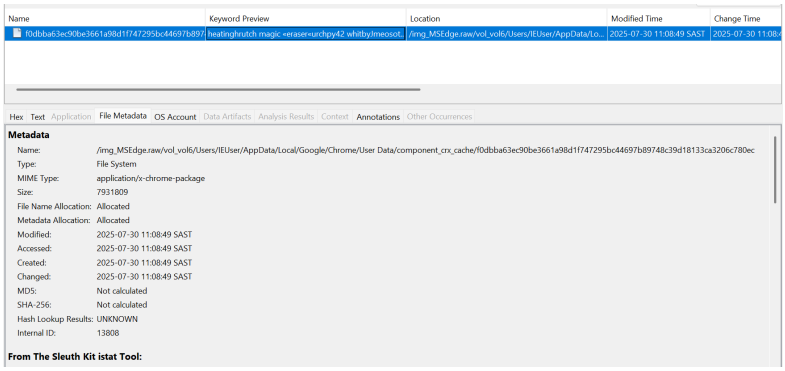


Figure 19: ”eraser” Key Search Output1

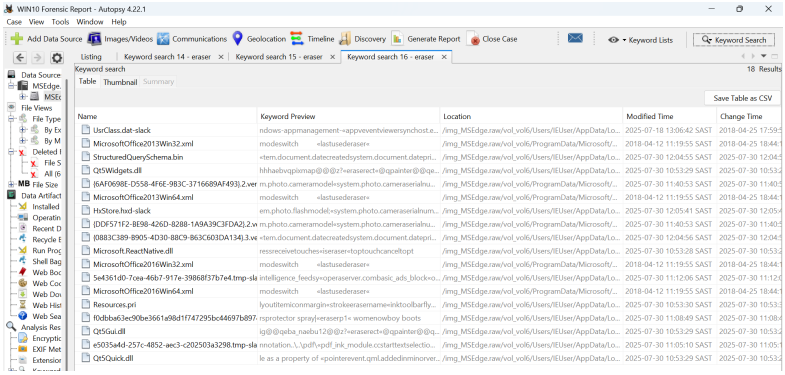


Figure 20: ”eraser” Key Search Output2

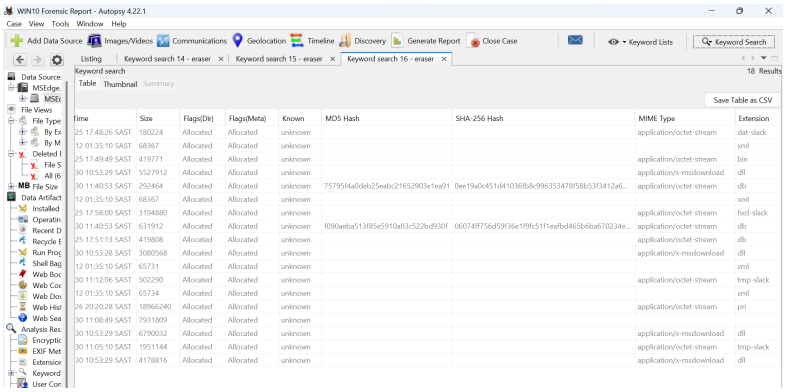


Figure 21: ”eraser” Key Search Output3

Suspicious activity began on **2018-01-25** with anti-forensic behavior, escalated through malware propagation in August 2018, and resurfaced in July 2025 with

coordinated downloads, deletion, and privacy tool usage. The timeline reflects both infection and intentional concealment.

Task 4. Browser and Network Activity

Browser artifacts

4.1. Browser History (Saved table as .csv, then used AI to analyze it)

Browser artifacts were extracted from `WebCacheV01.dat`, confirming that Microsoft Edge was the primary browser in use. The user's activity includes login attempts to Microsoft services via `live.com`, Bing search queries related to torrenting and VPNs, and direct access to `.torrent` files and suspicious HTML content. Downloads from OpenVPN, NordVPN, and Windscribe were also identified, alongside visits to GitHub pages referencing malware, specifically the EternalRocks worm, which exploits SMB vulnerabilities.

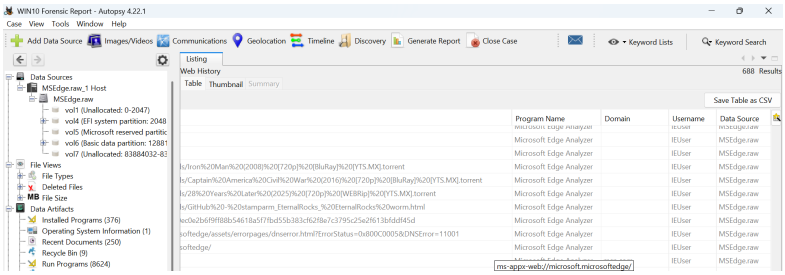


Figure 22: Web History Output

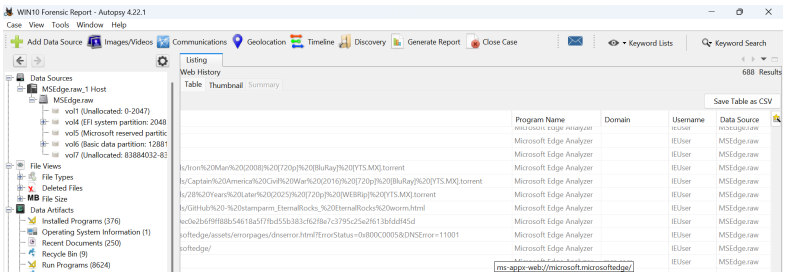


Figure 23: Web History - No domains

4.2. Web Downloads (Saved table as .csv, then used AI to analyze it)

Local file access logs show `.torrent` downloads for movies such as Iron Man (2008), Captain America: Civil War (2016), and 28 Years Later (2025), all sourced

from YTS.mx, a known torrenting platform. Additionally, the user accessed a GitHub-hosted HTML file referencing EternalRocks, indicating interest in or malware research. VPN-related activity was confirmed through searches and downloads from `openvpn.net`, `windscribe.com`, and `nordvpn.com`, suggesting efforts to anonymize browsing, likely to avoid detection while torrenting or accessing restricted content.

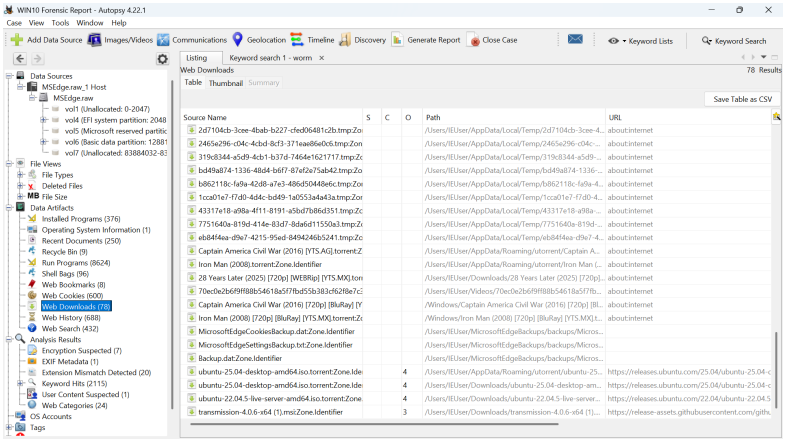


Figure 24: Web Downloads Output

4.3. Browser Cookies (Saved table as .csv, then used AI to analyze it)

Although no direct IP addresses were recovered, several artifacts imply outbound network activity. These include the download of a VPN client installer (`openvpn-connect-3.7.3.4351_signed.msi`), Bing searches such as "how to torrent without getting caught," and DNS error pages (`dnserror.html`) that suggest failed or redirected network attempts. These behaviors collectively point to attempts at anonymization and possible efforts to bypass network restrictions.

Cookies extracted from `WebCacheV01.dat` show consistent interaction with domains such as `bing.com`, `msn.com`, `live.com`, `youtube.com`, `linkedin.com`, and `google.com`, reflecting general browsing, search activity, and Microsoft account usage. More notably, cookies from `nordvpn.com` and `windscribe.com` confirm active VPN usage through session identifiers and referral metadata. These VPN services are commonly used to mask identity during torrenting or forum access. No cookies were found from known hacking forums such as `nulled.to`, `exploit.in`, or `breachforums.is`. However, indirect indicators, including GitHub malware references, torrenting behavior, and proxy artifacts such as `localhost:57585/proxy/0/`, suggest intent to anonymize and possibly engage

in suspicious activity.

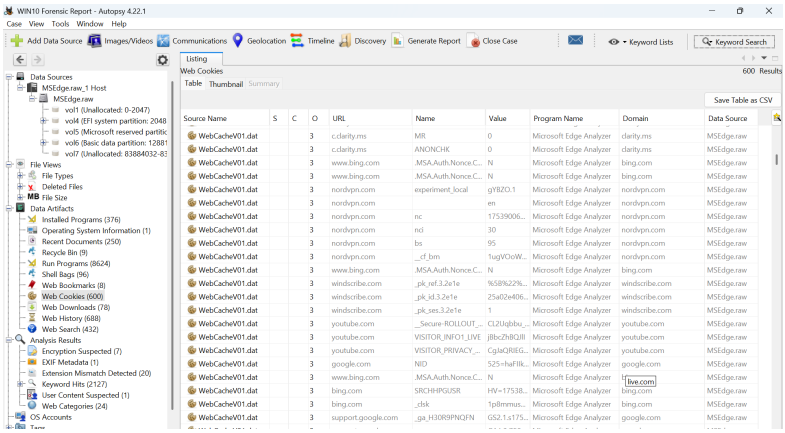


Figure 25: Browser Cookies Output

Task 5: Email or Messaging Artifact Recovery

1. Skype Filters

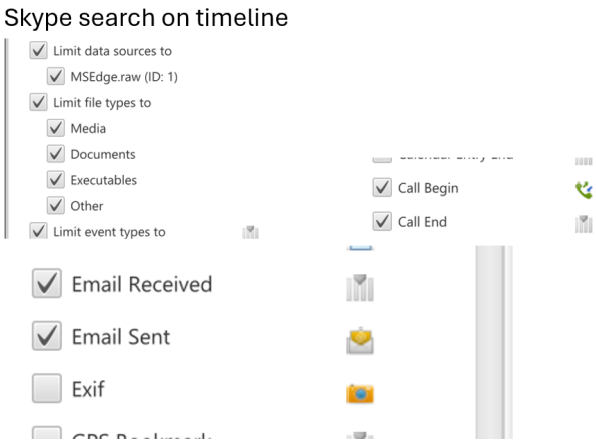


Figure 26: Skype filters chosen during artifact recovery

2. Skype Timeline and Usage

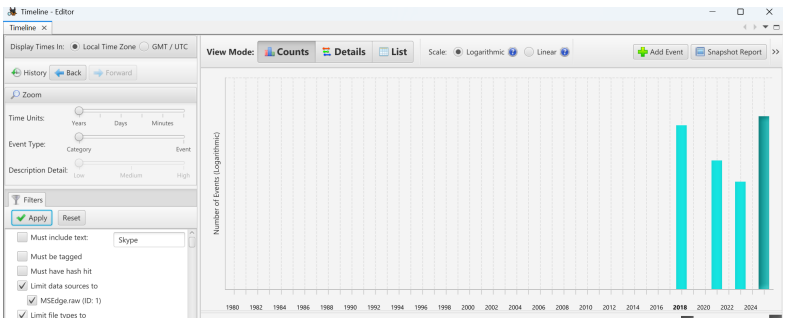


Figure 27: Skype use timeline output

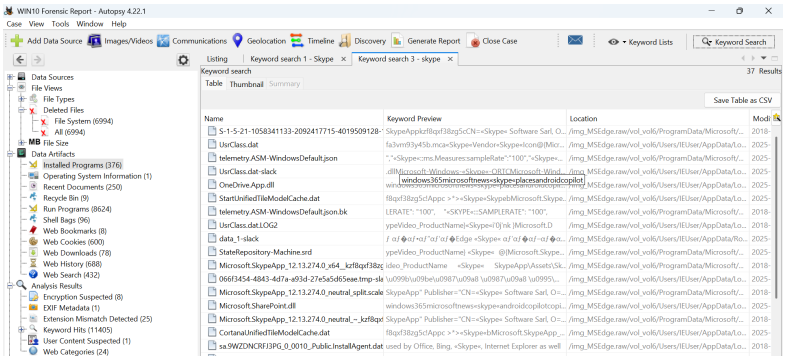


Figure 28: Evidence of Skype usage

3. Thunderbird

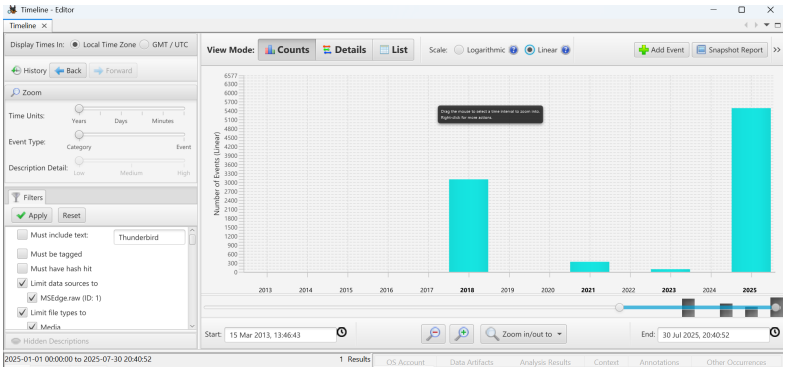


Figure 29: Thunderbird use timeline output

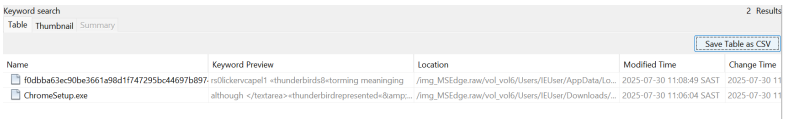


Figure 30: Evidence of Thunderbird usage

4. Outlook

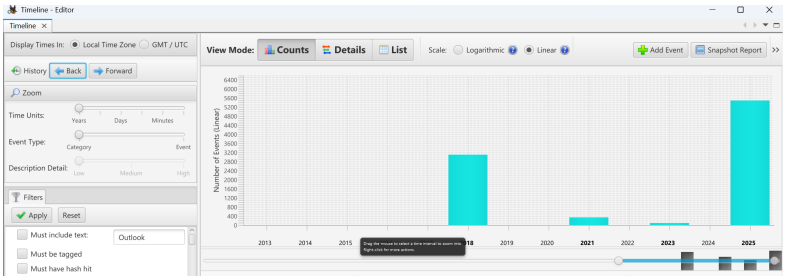


Figure 31: Outlook use timeline output

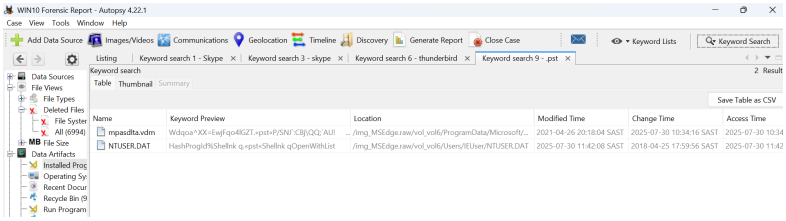


Figure 32: Evidence of Outlook usage via ".pst" key search

5. Suspicious Emails

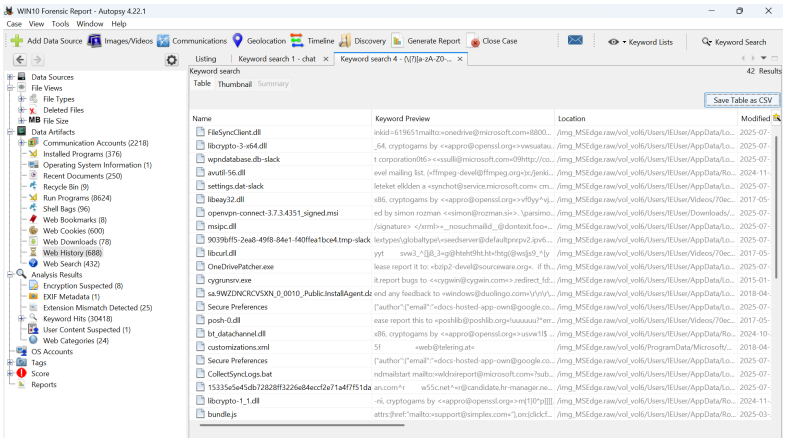


Figure 33: Suspicious email evidence via regular expression

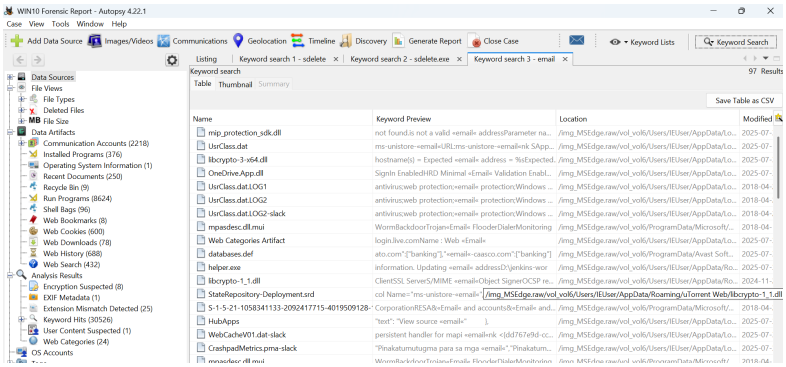


Figure 34: Suspicious email evidence via ”email” key search

Task 6: Suspicious Software or Malware Evidence

6.1. Suspicious Executables or Tools

Figure 35: Suspicious executables identified using ”trace” key search

6.2. Hash Verification and Threat Analysis

Figure 36: SHA-256 Hash: 28c03d41ec28365f9cb58c1281b29d7e6ca64decf77e2ba52807f4a4914e038a

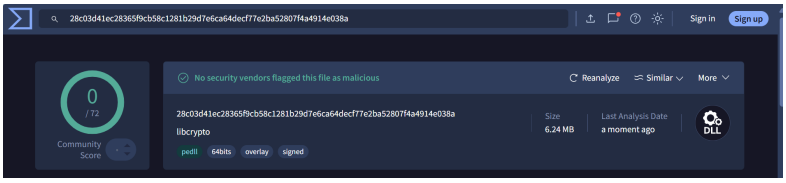


Figure 37: Threat analysis result: No security vendors flagged this file

Conclusion and Recommendations

The analysis revealed the presence of suspicious executables, VPN activity, and browser traces linked to torrenting and malware. Registry entries and startup

configurations indicated attempts to maintain persistence and conceal digital footprints. Based on these findings, the case should be escalated for disciplinary review. The compromised system must be reimaged, and stricter network and software policies should be enforced. Additionally, cybersecurity awareness training is recommended to prevent future incidents. These actions will strengthen digital oversight and uphold institutional integrity.